Assignment 4 - Task 1

Sequential Handwritten Digit Classification using Recurrent Neural Networks

Item	Details
Dataset	Dig-MNIST.csv - 10,240 labelled 28 x 28 Kannada digit images
Input representation	28 timesteps x 28 features; each image row is one timestep
Models	Vanilla RNN, LSTM and GRU
Configurations	12: 3 cell types x 2 depths x 2 hidden sizes
Optimizer	Adam (learning rate 0.001)
Loss	Cross-entropy
Model selection	Highest validation accuracy

1. Objective

The experiment classifies handwritten Kannada digits by converting each 28 x 28 image into a sequence of 28 row vectors. Vanilla RNN, LSTM and GRU models are compared using different recurrent depths and hidden sizes.

2. Dataset and Preprocessing

The notebook uses Dig-MNIST.csv containing 10,240 labelled 28 x 28 Kannada digit images. Pixel values are normalized to [0,1]. The split contains 7,372 training samples, 820 validation samples and 2,048 held-out test samples.

Sample Kannada-MNIST Images

Five examples are shown for each of the ten digit classes.



Figure 1. Five sample Kannada digit images from each of the ten classes.

3. Model and Experimental Setup

A recurrent classifier receives the 28 row vectors sequentially. The final recurrent output is passed to a linear layer with 10 outputs, one for each digit class.

Cell type	Layers	Hidden units
RNN	1	64
RNN	1	128
RNN	2	64
RNN	2	128
LSTM	1	64
LSTM	1	128
LSTM	2	64
LSTM	2	128
GRU	1	64
GRU	1	128
GRU	2	64
GRU	2	128

Training details

Adam optimization was used with learning rate 0.001. Gradient clipping at 5.0 was applied. The loss-difference convergence test used a threshold of |Lbar(k) - Lbar(k-1)| < 1e-4, with 30 epochs serving as the recorded safety cap in these runs.

Selection rule

Validation accuracy was used to select the final configuration. The held-out test set was used for final evaluation and comparative reporting.

4. Training Error and Gradient Flow

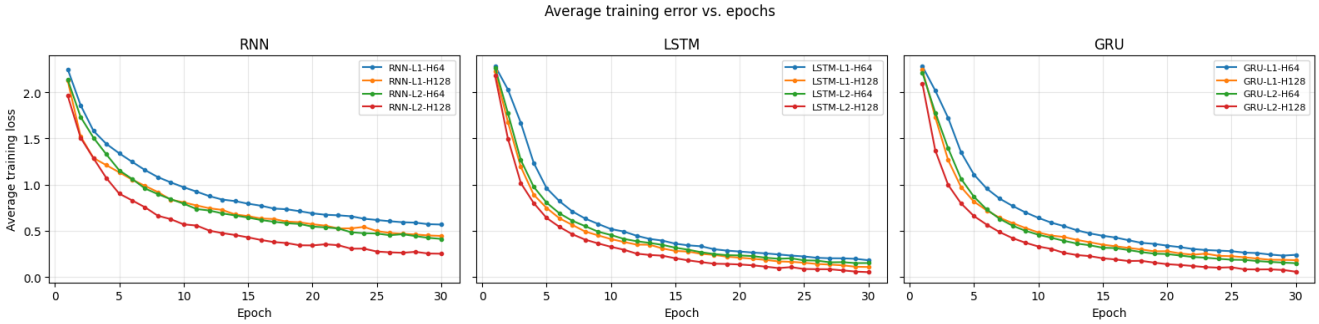


Figure 2. Average training error versus epoch for all twelve configurations.

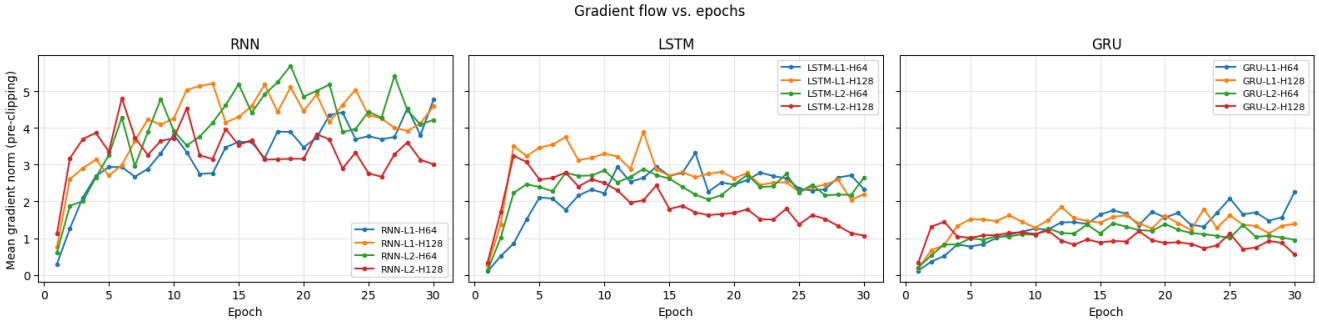


Figure 3. Mean gradient norm versus epoch for all twelve configurations.

The training-loss curves decrease across the runs. Larger hidden sizes and deeper recurrent models generally achieve lower final loss. The RNN curves remain above the gated models, while GRU-L2-H128 reaches the lowest final loss among the tested configurations.

5. Classification Results

Validation accuracy is the model-selection criterion, while the held-out test set provides the final evaluation.

Config	Cell	L	H	Ep.	Loss	Grad.	Train	Val	Test
RNN-L1-H64	RNN	1	64	30	0.5671	3.3132	82.62%	79.39%	80.08%
RNN-L1-H128	RNN	1	128	30	0.4440	4.0951	86.42%	82.68%	83.35%
RNN-L2-H64	RNN	2	64	30	0.4120	4.0511	88.33%	85.37%	85.06%
RNN-L2-H128	RNN	2	128	30	0.2515	3.3759	92.85%	88.66%	88.43%
LSTM-L1-H64	LSTM	1	64	30	0.1805	2.2683	94.26%	88.90%	88.82%
LSTM-L1-H128	LSTM	1	128	30	0.1074	2.7370	96.62%	90.49%	91.21%
LSTM-L2-H64	LSTM	2	64	30	0.1505	2.3437	95.09%	89.15%	89.45%
LSTM-L2-H128	LSTM	2	128	30	0.0517	1.9203	99.28%	91.22%	92.77%
GRU-L1-H64	GRU	1	64	30	0.2392	1.3308	93.64%	88.41%	89.79%
GRU-L1-H128	GRU	1	128	30	0.1818	1.3646	95.33%	91.10%	90.72%
GRU-L2-H64	GRU	2	64	30	0.1488	1.0718	95.77%	90.49%	91.26%
GRU-L2-H128	GRU	2	128	30	0.0569	0.9506	98.94%	93.66%	93.65%

Table 2. Results for all twelve recurrent-network configurations.

Average results by cell type

Cell	Final loss	Mean grad. norm	Train acc.	Test acc.
RNN	0.4186	3.7088	87.56%	84.23%
LSTM	0.1225	2.3173	96.31%	90.56%
GRU	0.1567	1.1794	95.92%	91.36%

Table 3. Mean results grouped by recurrent cell type.

6. Best Configuration

Selected model: GRU-L2-H128, chosen using the highest validation accuracy.

Metric	Value
Training accuracy	98.94%
Validation accuracy	93.66%
Test accuracy	93.65%
Final training loss	0.0569
Mean gradient norm	0.9506
Test samples	2,048

Table 4. Performance of the selected GRU-L2-H128 model.

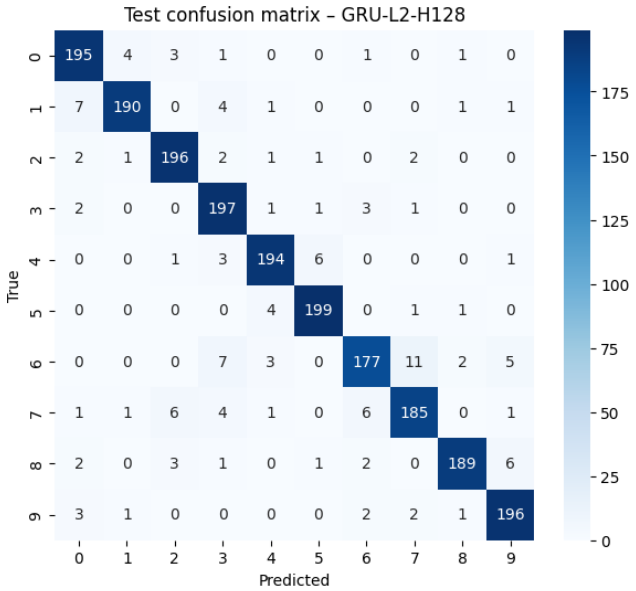


Figure 4. Confusion matrix of the selected GRU-L2-H128 model on the held-out test set.

7. Confusion Matrix and Per-Class Performance

True \ Pred	0	1	2	3	4	5	6	7	8	9
0	195	4	3	1	0	0	1	0	1	0
1	7	190	0	4	1	0	0	0	1	1
2	2	1	196	2	1	1	0	2	0	0
3	2	0	0	197	1	1	3	1	0	0
4	0	0	1	3	194	6	0	0	1	
5	0	0	0	0	4	199	0	1	1	0
6	0	0	0	7	3	0	177	11	2	5
7	1	1	6	4	1	0	6	185	0	1
8	2	0	3	1	0	1	2	0	189	6
9	3	1	0	0	0	2	2	1	196	

Table 5. Complete 10 x 10 confusion matrix for the selected model.

Digit	Precision	Recall	F1-score	Support
0	0.9198	0.9512	0.9353	205
1	0.9645	0.9314	0.9476	204
2	0.9378	0.9561	0.9469	205
3	0.8995	0.9610	0.9292	205
4	0.9463	0.9463	0.9463	205
5	0.9567	0.9707	0.9637	205
6	0.9267	0.8634	0.8939	205
7	0.9158	0.9024	0.9091	205
8	0.9692	0.9265	0.9474	204
9	0.9333	0.9561	0.9446	205

Table 6. Per-class precision, recall and F1-score for the selected model.

8. Observations and Comparative Analysis

• Vanilla RNN gives the weakest average test accuracy (84.23%).
• GRU gives the highest average test accuracy (91.36%), with LSTM close at 90.56%.
• Hidden size 128 and two recurrent layers generally improve performance.
• GRU-L2-H128 is the strongest configuration by validation accuracy (93.66%) and test accuracy (93.65%).
• The selected model has a training-test accuracy gap of about 5.29 percentage points.
• Digit 5 has the highest recall (97.07%), while digit 6 has the lowest recall (86.34%).

9. Convergence Note

The notebook includes the required loss-difference convergence test |Lbar(k) - Lbar(k-1)| < 1e-4. All twelve recorded runs reached the 30-epoch safety cap, so 30 epochs should be interpreted as the run limit rather than proof that every model satisfied the threshold.

10. Conclusion

The experiment compares Vanilla RNN, LSTM and GRU models for 28-step Kannada digit sequences. GRU-L2-H128 provides the best recorded validation and test performance, achieving 98.94% training accuracy, 93.66% validation accuracy and 93.65% test accuracy.